Original Inflation Calculation with Leontief

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May 21, 2026

Purpose: I want to derive an approximation that will connect the Leontief ShapirIO decomposition to net inflation defined the way Shapiro does it.

Let's get started.

1 Derivations*

CPI inflation is now calculated as:

$$\pi_{CPI,t} \equiv \sum_p \omega_{PCE,p,t} \underbrace{\left(\frac{P_{p,t}}{P_{p,t-1}} - 1\right)}_{\equiv \pi_{p,t}} = \underline{\omega_{PCE,t}}' \underline{\pi}_t$$

Note that our I-O decomposition has the following decomposition

$$\underline{P}_t = \underbrace{(I - A_t)^{-1} diag(\mathbb{1}_p' A_t)}_{\equiv B_t} \underline{P}_{VA,t}$$

where I put a t subscript on A_t because we have changes in the set of available products.

In summation terms this is:

$$P_{p,t} = \sum_{p'} B_{p,p',t} P_{VA,p',t}$$

Now we can define product level inflation as:

$$\pi_{p,t} \equiv \frac{P_{p,t} - P_{p,t-1}}{P_{p,t-1}} = \frac{\sum_{p'} \left(B_{p,p',t}(P_{VA,p',t} - P_{VA,p',t-1}) + (B_{p,p',t} - B_{p,p',t-1}) P_{VA,p,t-1} \right)}{\sum_{p''} B_{p,p'',t-1} P_{VA,p'',t-1}}$$

where in the numerator I added and subtracted $B_{p,p',t} P_{VA,p,t-1}$ to get this expression. Next we can do the following adjustments as well:

$$\pi_{p,t} = \frac{\sum_{p'} \left(B_{p,p',t}(P_{VA,p',t} - P_{VA,p',t-1}) \frac{P_{VA,p',t-1}}{P_{VA,p',t-1}} + (B_{p,p',t} - B_{p,p',t-1}) P_{VA,p,t-1} \frac{B_{p,p',t}}{B_{p,p',t}} \right)}{\sum_{p''} B_{p,p'',t-1} P_{VA,p'',t-1}}$$

$$= \frac{\sum_{p'} \left(B_{p,p',t} P_{VA,p',t-1} \frac{P_{VA,p',t} - P_{VA,p',t-1}}{P_{VA,p',t-1}} + B_{p,p',t} P_{VA,p,t-1} \frac{B_{p,p',t} - B_{p,p',t-1}}{B_{p,p',t}} \right)}{\sum_{p''} B_{p,p'',t-1} P_{VA,p'',t-1}}$$

$$= \underbrace{\sum_{p'} \frac{B_{p,p',t} P_{VA,p',t-1}}{\sum_{p''} B_{p,p'',t-1} P_{VA,p'',t-1}}}_{\equiv \omega_{VA,p,p',t}} \left(\pi_{VA,p',t-1} + \Delta B_{p,p',t} \right)$$

where $\pi_{VA,p',t} \equiv \frac{P_{VA,p',t} - P_{VA,p',t-1}}{P_{VA,p',t-1}}$ and $\Delta B_{p,p',t} \equiv \frac{B_{p,p',t} - B_{p,p',t-1}}{B_{p,p',t}}$

So far this is an exact decomposition (don't forget the time subscript in the change in B matrix – we divide by B in time t not $t-1$.

Now let's link this to CPI inflation:

$$\pi_{CPI,t} = \sum_{p} \omega_{PCE,p,t} \pi_{p,t}$$

$$= \sum_{p} \omega_{PCE,p,t} \sum_{p'} \omega_{VA,p,p',t} \left(\pi_{VA,p',t-1} + \Delta B_{p,p',t} \right)$$

$$= \sum_{p'} \sum_{p} \omega_{PCE,p,t} \omega_{VA,p,p',t} \left(\pi_{VA,p',t-1} + \Delta B_{p,p',t} \right)$$

$$= \sum_{p'} \left(\underbrace{\sum_{p} \omega_{PCE,p,t} \omega_{VA,p,p',t}}_{\equiv W_{p',t}} \pi_{VA,p',t-1} + \underbrace{\sum_{p} \omega_{PCE,p,t} \omega_{VA,p,p',t} \Delta B_{p,p',t}}_{\equiv V_{p',t}} \right) = \sum_{p'} W_{p',t} \pi_{VA,p',t-1} + V_{p',t}$$

And this is also an exact decomposition!

Implementation Steps

1. Start by constructing the time varying matrices $B_t \equiv (I - A_t)^{-1} diag(\mathbb{1}_p' A_t)$ which should be a different $\mathcal{P}_t$ x $\mathcal{P}_t$ for each time period (where $\mathcal{P}_t$ is the number of products in that period). Note also that this only differs by the set of products – if that set doesn't change, then $B_t = B_{t-1}$.

2. Then use this to construct $\omega_{VA,p,p',t} \equiv \frac{B_{p,p',t} P_{VA,p',t-1}}{\sum_{p''} B_{p,p'',t} P_{VA,p'',t-1}}$. Note that this is a different $\mathcal{P}_t$ x $\mathcal{P}_t$ matrix for each time period t.

3. Calculate value added inflation and the change in B matrices (not the denominator in the B formula):

$$\pi_{VA,p',t} \equiv \frac{P_{VA,p',t} - P_{VA,p',t-1}}{P_{VA,p',t-1}}$$
$$\Delta B_{p,p',t} \equiv \frac{\Delta B_{p,p',t} -]\Delta B_{p,p',t-1}}{\Delta B_{p,p',t}}$$

4. Then create $W_{p',t}$ and $V_{p',t}$

$$W_{p',t} \equiv \sum_{p} \omega_{PCE,p,t} \omega_{VA,p,p',t}$$
$$V_{p',t} \equiv \sum_{p} \omega_{PCE,p,t} \omega_{VA,p,p',t} \Delta B_{p,p',t}$$

5. Now with these check that the following holds:

$$\pi_{CPI,t} = \sum_{p'} \left(W'_{p',t} \pi_{VA,p',t} + V_{p',t} \right)$$

6. If not, let Nick know!

7. Then calculate demand and supply inflation as follows:

$$\pi^{Shapiro}_{demand,t} \equiv \sum_{p'} D_{p',t} \omega_{PCE,p',t} \pi_{p',t}$$
$$\pi^{Shapiro}_{supply,t} \equiv \sum_{p'} (1 - D_{p',t}) \omega_{PCE,p',t} \pi_{p',t}$$
$$\pi^{ShapirIO}_{demand,t} \equiv \sum_{p'} D_{p',t} \left(W'_{p',t} \pi_{VA,p',t} + V_{p',t} \right)$$
$$\pi^{ShapirIO}_{supply,t} \equiv \sum_{p'} (1 - D_{p',t}) \left(W'_{p',t} \pi_{VA,p',t} + V_{p',t} \right)$$

8. Do the 12 month averaging thing you already do and then the standard plots.